



Optimizing Polynomial and Regularization Techniques for Enhanced Housing Price Prediction Accuracy

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Abstract

This study investigates the effectiveness of various regression models for predicting housing prices using the California Housing dataset. The models evaluated include Linear Regression, Ridge Regression, Best Polynomial Ridge Regression, Lasso Regression, Elastic Net Regression, and Support Vector Regression (SVR) with an RBF kernel. The analysis reveals that SVR with an RBF kernel exhibits the poorest performance, characterized by the highest Mean Squared Error (MSE) and the lowest R^2 score, indicating limited effectiveness for this dataset. Conversely, Linear Regression, Ridge Regression, and Best Polynomial Ridge Regression demonstrate significantly lower MSE values and nearly identical R^2 scores, each explaining approximately 60% of the variance in housing prices. Among these, Best Polynomial Ridge Regression marginally outperforms Linear and Ridge Regression, suggesting that including polynomial features enhances model performance. Although Lasso Regression shows slightly higher MSE than the leading models, it still performs better than Elastic Net and SVR. Overall, the study identifies linear regression, ridge regression, and best polynomial ridge regression as the most effective models for this dataset, with SVR and an RBF kernel being the least effective.

Keywords Regression techniques · Linear regression · Lasso regression · Elastic net regression · Polynomial Ridge regression · SVR with RBF regression

Introduction

Regression analysis is a powerful statistical tool to explore relationships between variables, predict outcomes, and inform decision-making across various fields. It encompasses multiple types, including linear, logistic, and Poisson regression, each serving distinct purposes in data analysis and interpretation. Figure 1. Depicts various Regression analysis techniques. Linear Regression evaluates the relationship between a dependent variable and one or more independent variables to predict outcomes. Logistic Regression is used for binary outcomes, estimating the probability of an event occurring. Poisson Regression models count data, predicting the number of events within a fixed interval [1]. Regression models are widely used in various fields to improve decision-making and efficiency. Inventory management, they help optimize the Economic Order Quantity and Reorder Point, thus enhancing logistical efficiency [2]. In healthcare, regression analysis is increasingly applied in clinical research to identify risk factors and associations, although it is important to note that such models do not establish causation.

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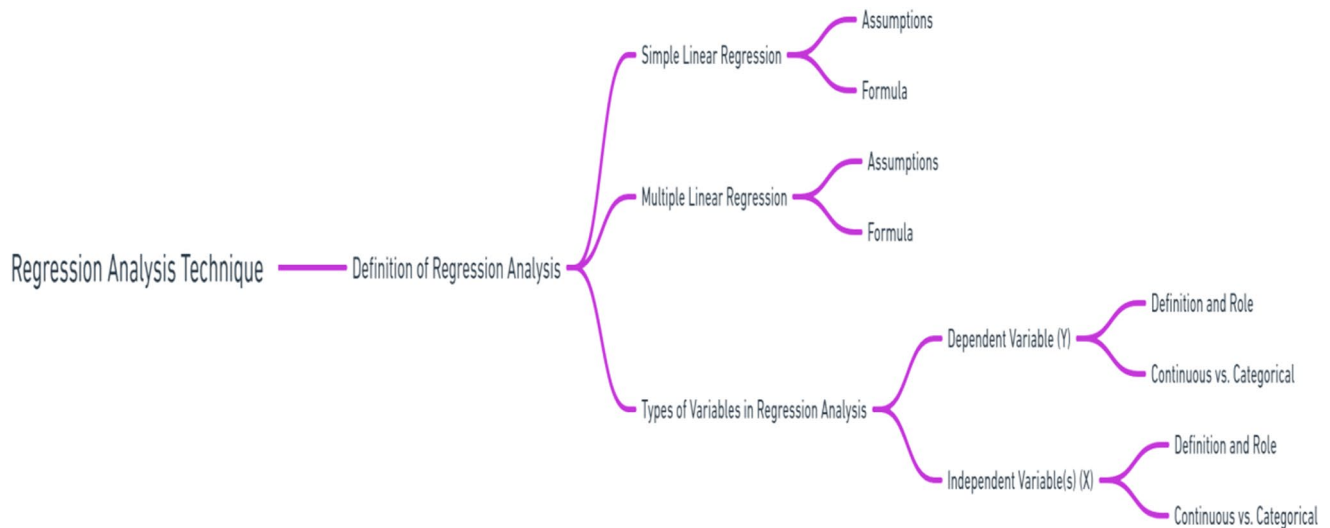


Fig. 1 Regression analysis techniques

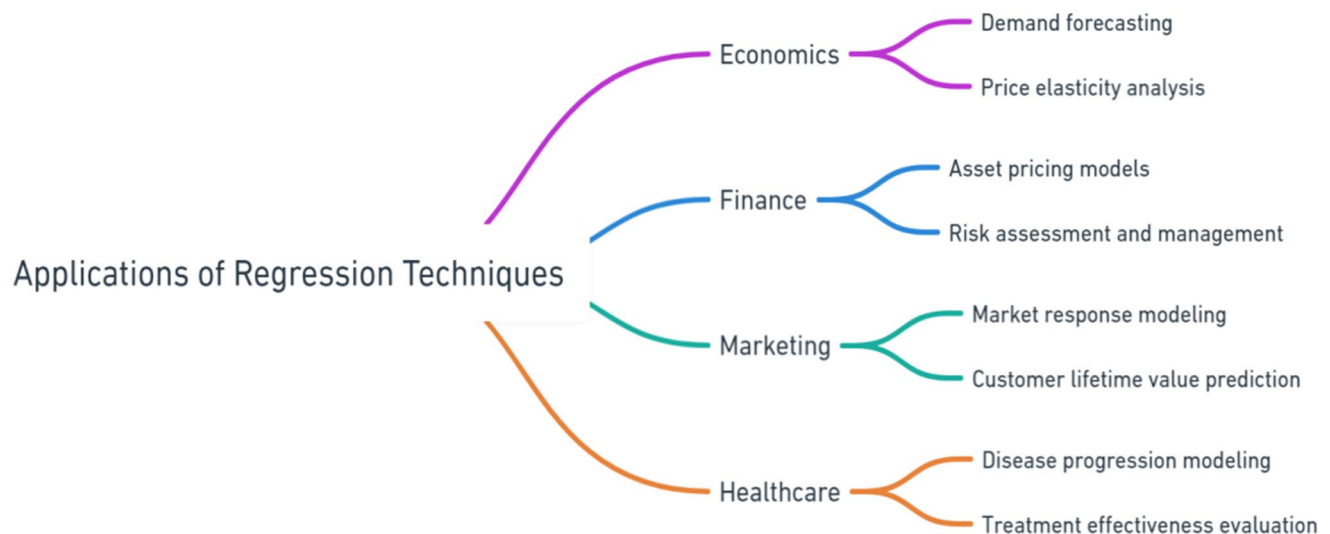


Fig. 2 Applications of regression techniques

Although regression analysis is crucial for deciphering complex data relationships, it is essential to acknowledge its limitations, especially concerning causation and the underlying assumptions. Critics argue that an exclusive focus on regression may miss broader developmental changes, indicating that a more nuanced approach might be necessary in some contexts [3]. Regression models can often misrepresent causal relationships. Even if a model is well-specified, it may lead to incorrect conclusions if the assumption that all other variables remain constant is violated [4]. The notion that regression can definitively establish causation is contested by the complexities of real-world interactions, where multiple factors simultaneously influence outcomes [5]. Figure 2 displays various applications of Regression techniques in real-world interactions.

Predicting housing prices accurately is a pivotal challenge in real estate economics, influencing investment decisions, market analysis, and policy-making. As housing markets grow more complex, traditional linear models often fall short of capturing intricate relationships between features and prices. Recent advancements in machine learning have introduced various regression techniques that promise to enhance predictive accuracy. This study delves into the performance of several such methods, aiming to provide a comprehensive evaluation of their efficacy using the California Housing dataset.

Existing research has explored various regression methods, including linear models, regularization techniques, and more complex algorithms like Support Vector Regression (SVR). Linear Regression and its regularized counterparts, Ridge and Lasso Regression, have been widely studied for

their ability to handle overfitting and feature selection. Polynomial regression and hybrid models like Elastic Net have also been employed to address non-linearity in data. Despite this, a lack of comparative analysis consolidates these methods' performance in a single study.

This paper positions itself by evaluating and comparing Linear Regression, Ridge Regression, Best Polynomial Ridge Regression, Lasso Regression, Elastic Net Regression, and SVR with an RBF kernel. The research problem identifies which regression technique offers the best balance of bias and variance for predicting housing prices. The problem statement addresses the need for a comprehensive analysis to guide model selection in real-world applications. The main objectives of this study are as follows:

- Assess the performance of various regression techniques, including Linear Regression, Ridge Regression, Polynomial Ridge Regression, Lasso Regression, Elastic Net Regression, and SVR with RBF kernel, to determine which models offer the best predictive accuracy for housing prices.
- Investigate how incorporating polynomial features into Ridge Regression affects model performance, to understand the benefits of capturing non-linear relationships in the data.
- Use Grid Search to fine-tune hyperparameters in Lasso Regression, Elastic Net Regression, and SVR, and evaluate how these adjustments improve model performance.

Literature Review

The study [6] compared ten machine learning models on various datasets using different training strategies and evaluation metrics, with a deep LSTM neural network showing superior performance in regression tasks. The research paper [7] presents a regression-based approach for explicitly modeling heat and mass transfer in microchannel heat exchangers, enhancing accuracy and speed for heat pump applications. The study [8] evaluates regression models like SVR, KNN, Random Forest, Gradient Boosting, and XGBOOST for COVID-19 reproduction rate prediction, emphasizing feature selection and hyperparameter tuning impact on performance. Regression models accurately predict outlet conditions of counter-flow dew point evaporative cooling systems, aiding in system design and optimization with a high level of explained variability [9]. The study [10] compared eleven regression models using criteria measures like AIC, SIC, and HQIC. The exponential growth model outperformed others in analyzing baby weight and mothers' hemoglobin levels. The study [11] compared linear and polynomial regression models for predicting Indian GDP,

with the polynomial model showing superior accuracy in capturing non-linear relationships, surpassing the linear model's performance. The paper [12] compares LR models with an LP-based possibilistic model using real data, showing the best linear models for regression analysis on the dataset. The research [13] compares BP neural network, extreme learning machine, and support vector machine for regression, evaluating single and integrated models' performance on four datasets. The study [14] compared Linear, Ridge, Lasso, Decision Tree, and Support Vector Regression models for software effort estimation. Lasso Regression performed best with the lowest Mean Absolute Error. Other well-known regression models involve [15, 16].

Preliminaries

In this section, the theoretical background of the models used is explained.

Linear Regression

Linear Regression models the relationship between a dependent variable, y , and one or more independent variables, X , using a linear equation. The goal is to find the best-fit line that minimizes the difference between the predicted and actual values.

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon \quad (1)$$

Where y is the dependent variable, $X_1, X_2, \dots, X_p$ are independent variables, β_0 : Intercept, $\beta_1, \beta_2, \dots, \beta_p$ are Coefficients and ε : error term. The objective is to minimize the Mean Squared Error (MSE) between observed values and predictions.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y - \hat{y}_i)^2 \quad (2)$$

Ridge Regression

Ridge Regression is a regularized version of Linear Regression that includes a penalty term to prevent overfitting by shrinking the regression coefficients. The penalty term is proportional to the square of the magnitude of the coefficients. Ridge Regression adds an L2 regularization term to the cost function, which penalizes large coefficients to prevent overfitting. The cost function becomes:

$$\text{Function} = \frac{1}{n} \sum_{i=1}^n (y - \hat{y}_i)^2 + \lambda \sum_{j=1}^n \beta_j^2 \quad (3)$$

Regularization parameter controlling the strength of the penalty. When $\lambda = 0$, Ridge Regression reduces to ordinary Linear Regression.

Polynomial Ridge Regression

The input features x_i are transformed into polynomial features before applying Ridge Regression. The general form is:

$$\hat{y} = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \dots \quad (4)$$

Similar to Ridge Regression but applied to polynomial features. The cost function still includes the L2 regularization term:

$$\text{Function} = \frac{1}{n} \sum_{i=1}^n (y - \hat{y}_i)^2 + \lambda \sum_{j=1}^n \beta_j^2 \quad (5)$$

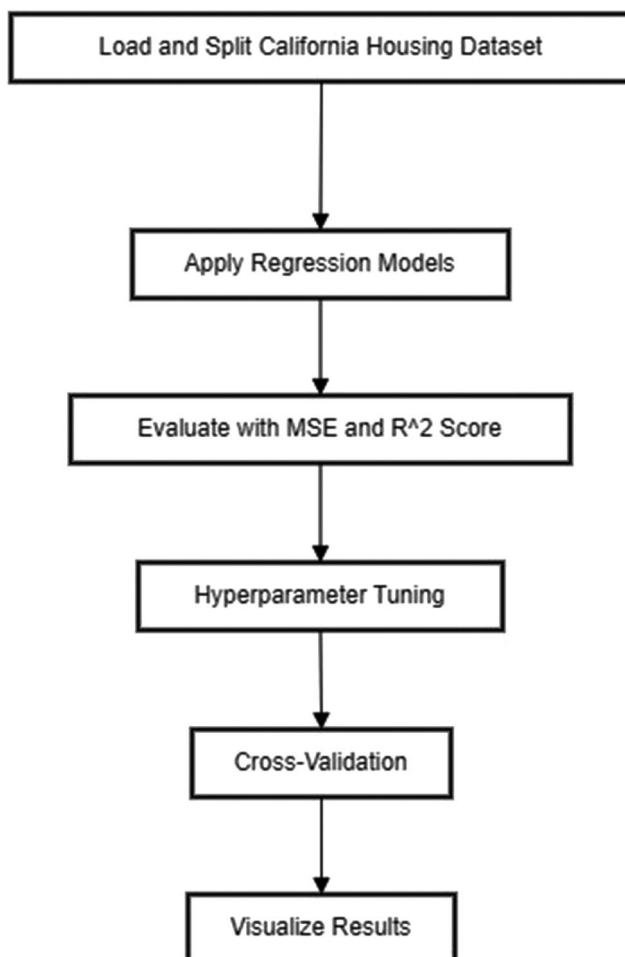


Fig. 3 Methodology followed

Lasso Regression (L1 Regularization)

Lasso Regression adds an L1 regularization term to the cost function, which encourages sparsity in the coefficients. The cost function is:

$$\text{Function} = \frac{1}{n} \sum_{i=1}^n (y - \hat{y}_i)^2 + \lambda \sum_{j=1}^n |\beta_j| \quad (6)$$

Unlike Ridge Regression, Lasso can shrink some coefficients to exactly zero, effectively selecting a subset of features. This makes Lasso useful for feature selection in high-dimensional datasets.

Elastic Net Regression

Elastic Net combines L1 and L2 regularization, adding both Lasso (L1) and Ridge (L2) penalties to the cost function:

$$\text{Function} = \frac{1}{n} \sum_{i=1}^n (y - \hat{y}_i)^2 + \lambda_1 \sum_{j=1}^n |\beta_j| + \lambda_2 \sum_{j=1}^n \beta_j^2 \quad (7)$$

λ_1 and λ_2 control the strength of the L1 and L2 penalties.

Support Vector Regression (SVR) with RBF Kernel

The general form of the SVR model is similar to that of linear models, but it optimizes a different objective function:

$$\hat{y} = \sum_{i=1}^n \alpha_i K(x_i, x) + b \quad (8)$$

Here, $K(x_i, x)$ is the kernel function that transforms the input data into a higher-dimensional space. The RBF (Radial Basis Function) kernel is one of the most used kernels:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (9)$$

SVR aims to find a hyperplane that best fits the data within a tolerance margin (ϵ). The optimization problem is:

$$\min \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j K(x_i, x_j) - \sum_{i=1}^n \alpha_i \quad (10)$$

Methodology

Figure 3 show the research methodology which involves a comprehensive evaluation of various regression models applied to the California Housing dataset. Initially, the dataset, which includes features like MedInc, HouseAge, and AveRooms, is loaded and split into training and testing sets. Six regression models are considered: Linear Regression,

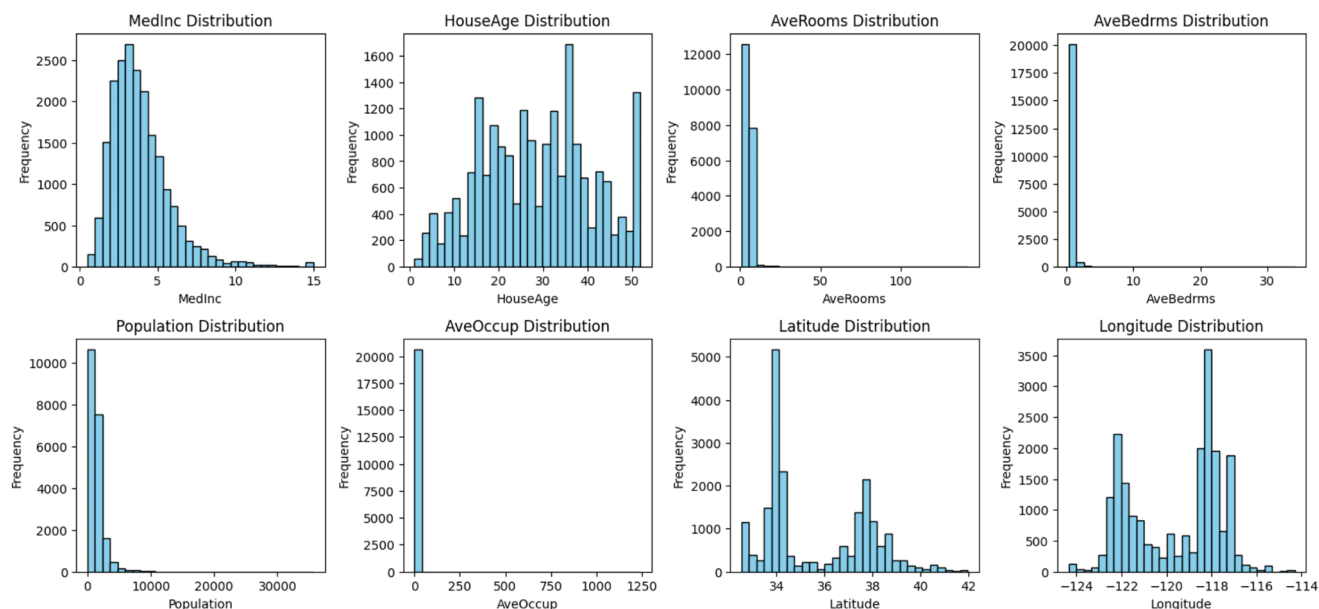


Fig. 4 Frequency distribution

Ridge Regression, Polynomial Ridge Regression, Lasso Regression, Elastic Net Regression, and Support Vector Regression (SVR). For each model, the performance is assessed using Mean Squared Error (MSE) and R^2 score. Polynomial Ridge Regression, Lasso Regression, Elastic Net Regression, and SVR are fine-tuned using GridSearchCV to optimize hyperparameters, including polynomial degree, ridge alpha, lasso alpha, elastic net alpha and l1_ratio, and SVR's C and epsilon. The models are evaluated through cross-validation with KFold to ensure robust performance metrics. Results are visualized through bar charts comparing MSE and R^2 scores, and scatter plots of true values versus predictions for the best-performing model. Additionally, learning curves for Linear and Ridge Regression are plotted to analyze performance as training data size varies, and residual plots are examined for the best model to assess fit quality. The analysis utilizes Python with libraries such as numpy, pandas, matplotlib, and scikit-learn for machine learning and visualization tasks.

Dataset Description

This section describes various characteristics of the dataset used in the study. Figure 4 illustrates frequency distribution of all the features.

Figure 4 displays eight histograms representing the distribution of various features in a dataset. Median income (MedInc) is right-skewed, with most values between 2 and 6 and a peak around 4. House age is more evenly distributed, peaking around 20–30 years. Average rooms (AveRooms) is heavily skewed, with most values under 10, while average bedrooms (AveBedrms) is sharply concentrated below

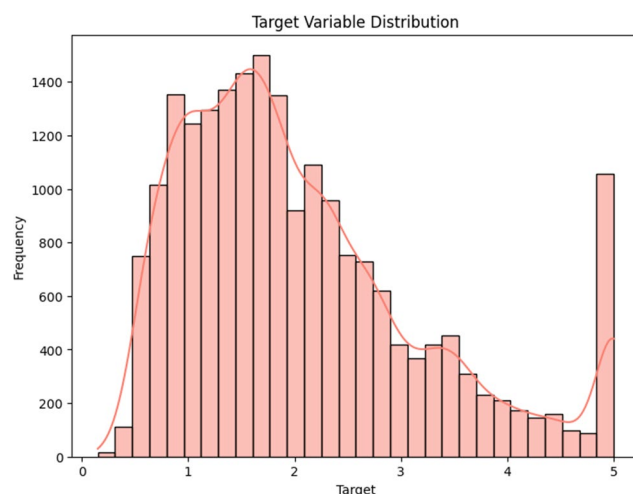


Fig. 5 Target variable distribution

5. The population distribution shows a strong left skew, with most values under 10,000. Average occupancy (AveOccup) also exhibits left skewness, peaking around 3. Latitude and longitude distributions show clusters, with latitude values mainly between 32 and 42, and longitude peaking near -118 . Figure 5 shows the distribution of a target variable, with values ranging from 0 to 5. The most frequent target values fall between 1 and 2, peaking at around 1.5 with a frequency slightly over 1,400. The distribution follows a unimodal pattern with a skew towards the right, showing a decrease in frequency after the peak, dropping below 200 for values greater than 4.

However, there is a notable spike around 5, with the frequency rising back up to approximately 600. Figure 6 depicts the heatmap of correlation among features.

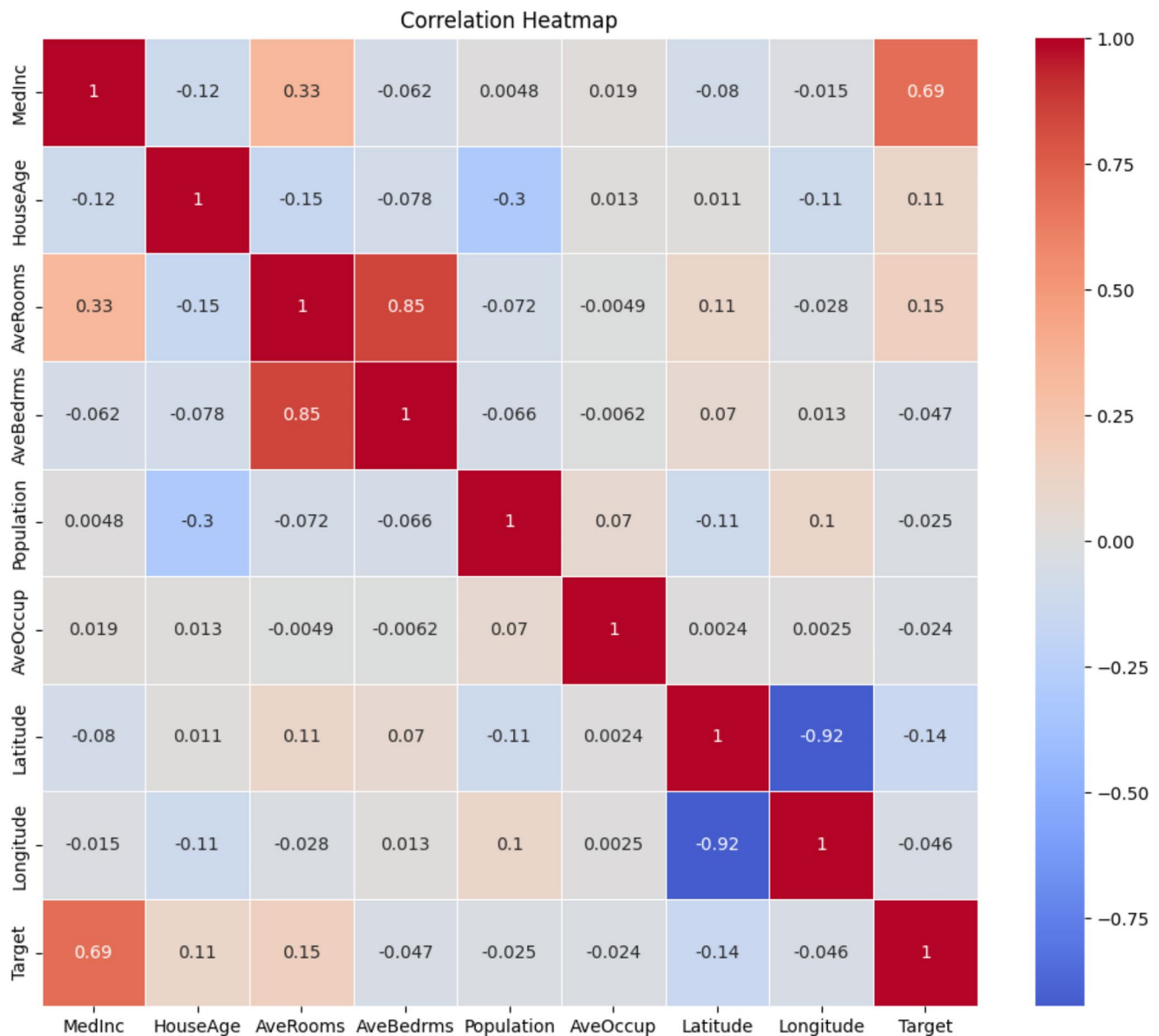


Fig. 6 Correlation heatmap

Results

This section illustrates performance evaluation results of all the models used in this study in terms of MSE, R^2 , True values vs. predictions, predicted vs. residuals and learning curves of all the models. Figure 7 illustrates MSE of all the models.

It is evident from Fig. 7 that Support Vector Regression (SVR) with RBF Kernel exhibits the highest Mean Squared Error (MSE), indicating its poor performance relative to the other models in the dataset. In contrast, Linear Regression, Ridge Regression, Best Polynomial Ridge Regression, and Elastic Net Regression demonstrate low and similar MSE values, suggesting comparable and better performance

among them. Lasso Regression, although showing a slightly higher MSE compared to Linear and Ridge Regression, still outperforms Elastic Net and SVR with RBF Kernel. Notably, Best Polynomial Ridge Regression marginally outshines Ridge and Linear Regression, suggesting that the inclusion of polynomial features has enhanced its performance. Overall, Linear Regression, Ridge Regression, and Best Polynomial Ridge Regression stand out as the top-performing models, while the SVR with RBF Kernel is notably less effective for this particular dataset. Figure 8 graphically illustrates comparison of R^2 values.

Figure 8 shows that Linear Regression, Ridge Regression, and Polynomial Ridge Regression achieve the highest and nearly identical R^2 scores, explaining about 60%

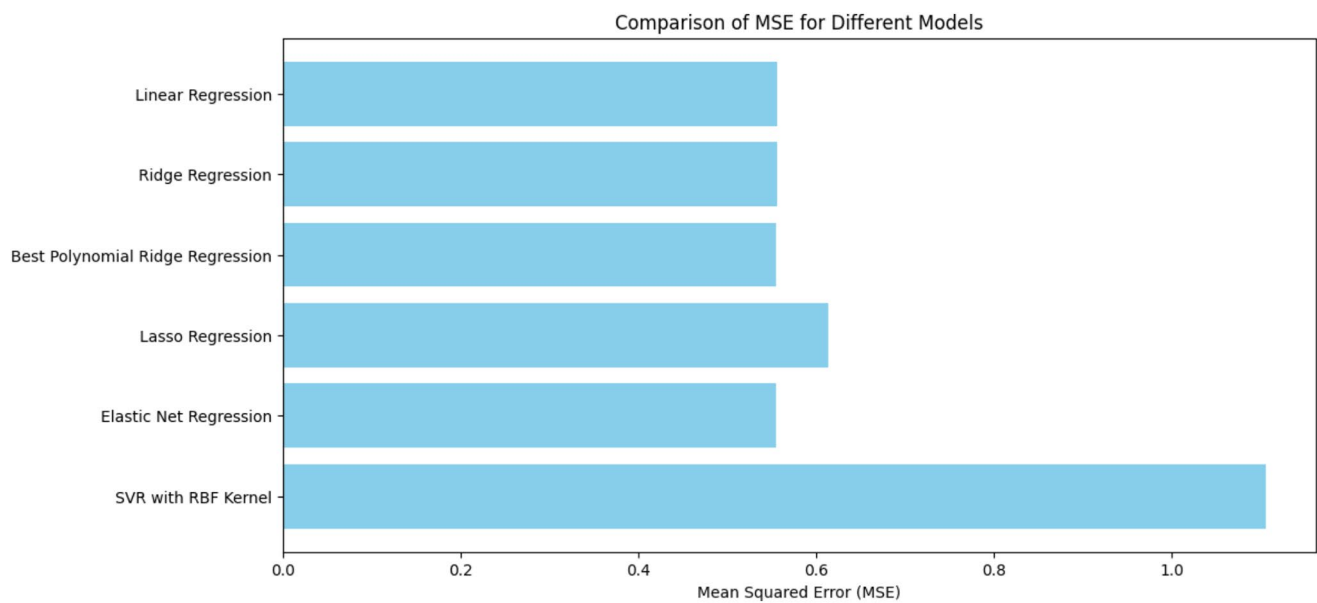


Fig. 7 Comparison of MSE

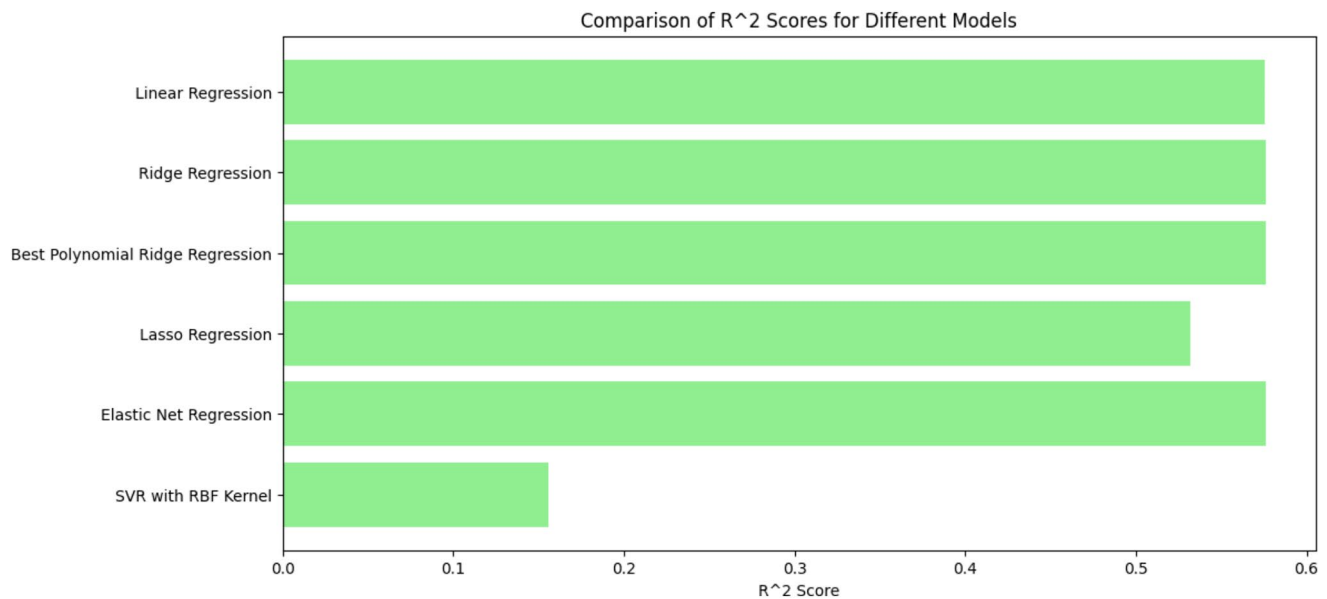


Fig. 8 Comparison of R^2

of the variance in the target variable, making them the best models. Elastic Net performs well but slightly lags behind. Lasso Regression has a lower R^2 score, while SVR with RBF Kernel performs the worst, explaining little variance. Figure 9 shows true vs. predicted values.

Linear Regression, Ridge Regression, and Polynomial Ridge Regression demonstrate the best performance, with predictions closely aligned with the true values. Elastic Net Regression and Lasso Regression show more variability, with Lasso particularly underpredicting higher values. SVR with RBF Kernel significantly underperforms, producing predictions that are almost flat and failing to capture the

range of true values effectively. Figure 10 illustrates predicted values vs. residual values.

Figure 10, compares six regression models through scatter plots, showing their predicted versus measured values. Linear and Ridge Regressions have closely packed data points along the diagonal, indicating good fits with some variance. Polynomial Ridge Regression captures more complex relationships with noticeable curvature. Lasso Regression shows a wider spread, suggesting less precision. Elastic Net Regression combines features of Lasso and Ridge, displaying a unique spread pattern. SVR with RBF Kernel handles most data well but has some outliers, indicating it

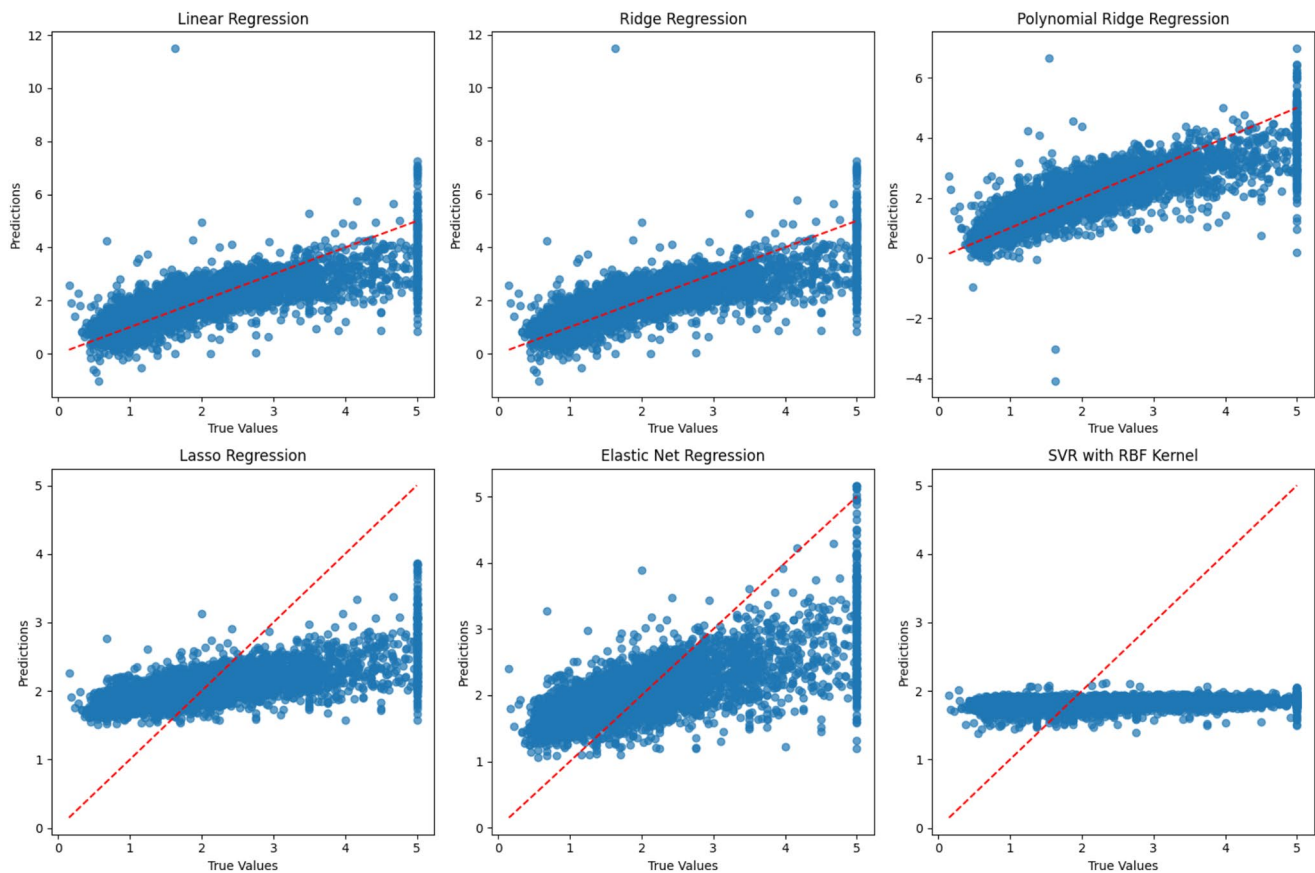


Fig. 9 True vs. predicted values

struggles with certain points. Overall, Polynomial Ridge Regression appears to capture the most complex relationships, while SVR with RBF Kernel has the most outliers. Figure 11 depicts learning curve for linear regression.

From Fig. 10 it can be inferred that as training size increases, the model's training error decreases, and validation error initially decreases before stabilizing, indicating improved performance with more data. The shaded areas show score variability, with both errors converging as data size grows. The graph highlights the importance of sufficient training data for better predictive accuracy. Figure 11 shows learning curve for Ridge regression.

Analysis from Fig. 11 shows that both training and validation scores start high and decrease with increasing training size, plateauing around 8000 examples. The MSE for both scores approaches zero as the training size grows. Variance decreases with larger training sizes, indicating more stable performance. The graph illustrates that performance improves with more data up to 8000 examples, after which additional data yields minimal gains. Figure 12 describes learning curve of polynomial Ridge regression.

From Fig. 12 it is evident that the training score, starting near 0 MSE, gradually improves as the training size increases, with variance decreasing and indicating more

stability. The validation score begins high at around 500,000 MSE, drops sharply, and then plateaus with increasing training size. The training size ranges from 0 to over 14,000, and MSE spans from $-100,000$ to $500,000$. Overall, the graph illustrates that while model performance improves with additional data, the gains become minimal beyond a certain point. Figure 13 displays learning curve for Lasso regression.

As inferred from Fig. 13 Initially, both training and validation scores improve with more data, but eventually worsen, suggesting overfitting. The training score drops sharply then slightly increases, while the validation score declines initially before rising again. Optimal performance appears around 4000 to 6000 training examples. The graph highlights that while more data initially benefits the model, excessive data can lead to overfitting, emphasizing the need to balance data size carefully. Figure 14 describes the learning curve of Elastic net regression.

As evidenced from Fig. 14 Initially, both training and validation scores improve with more data, but then both begin to worsen, indicating potential overfitting. The training score rises sharply before stabilizing, while the validation score decreases initially and then increases again. Optimal performance is seen with a training size of around

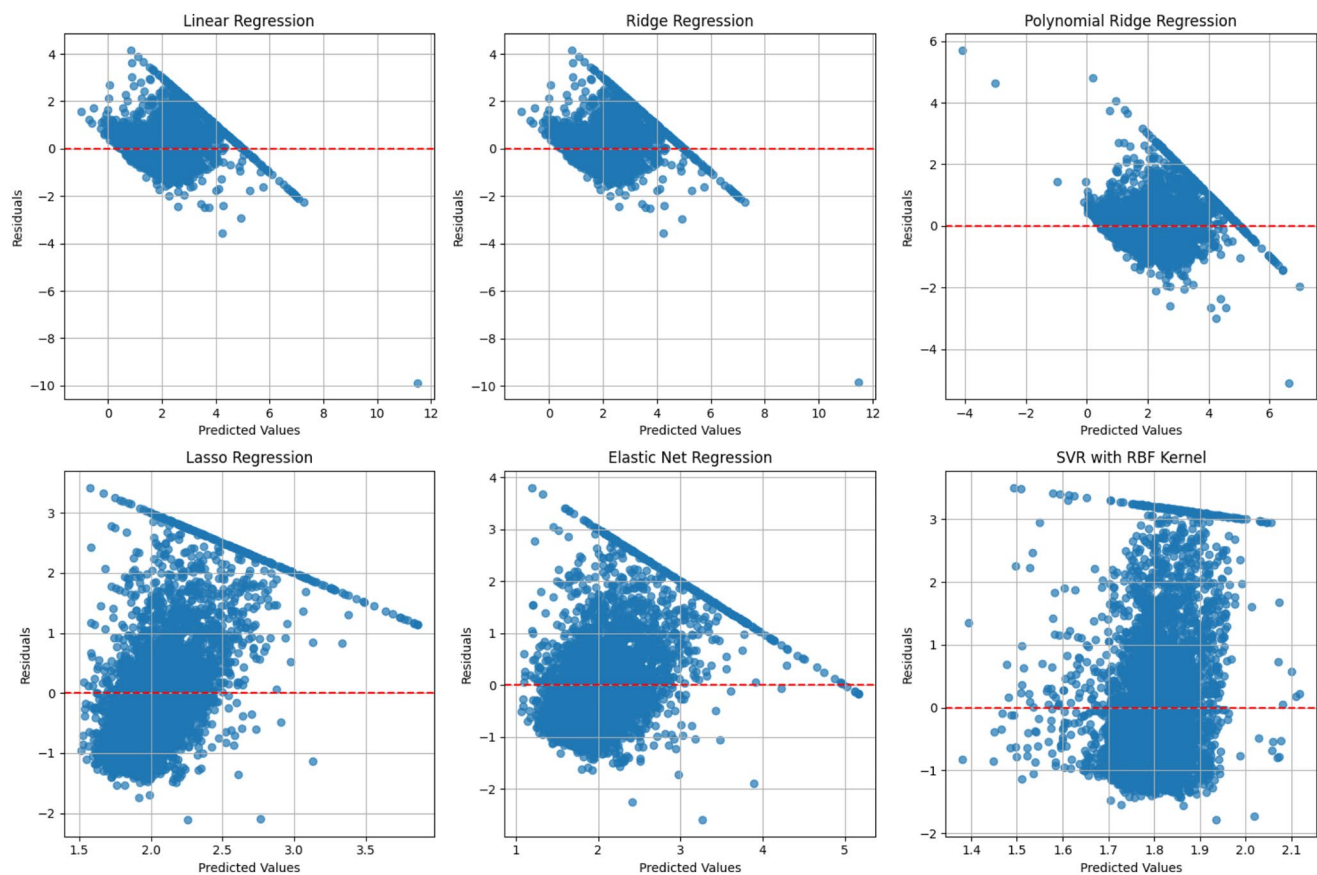


Fig. 10 Predicted vs. residual values

4000 to 6000. The graph highlights that while more data can enhance model performance initially, excessive data may lead to overfitting, underscoring the importance of balancing data size and monitoring both scores. Figure 15 depicts learning curve with respect to SVR with RBF kernel.

It is inferred from Fig. 15 Initially, both scores start high, indicating a higher error rate. The training score rises, suggesting potential overfitting, but then decreases and stabilizes, showing improved learning with more data. The validation score also peaks and then decreases, reflecting better performance on unseen data with increased training. Scores stabilize around a training size of 12,000, suggesting this is an optimal point for balancing bias and variance. The graph indicates that while initial training increases errors, performance improves and stabilizes with more data.

Conclusion

This study concludes that Linear Regression, Ridge Regression, and Best Polynomial Ridge Regression are the most effective models for predicting housing prices in the California Housing dataset, with each model explaining about 60% of the variance. The best Polynomial Ridge Regression

offers a slight performance edge by incorporating polynomial features. In contrast, SVR with an RBF kernel is identified as the least effective model, showing the highest Mean Squared Error (MSE) and the lowest R^2 score. The study's analysis is constrained to a limited set of regression models, and the performance is based solely on the California Housing dataset. Furthermore, the dataset's specific characteristics may limit the generalizability of the results to other housing markets or datasets. Feature engineering and hyperparameter tuning were not explored in depth, which may impact the models' performance. Future research could explore a broader range of machine learning models, including ensemble methods like Random Forest and Gradient Boosting. Enhancing feature engineering and performing hyperparameter optimization could further improve model accuracy. Additionally, testing the models on diverse datasets and incorporating more complex spatial and temporal features could enhance the generalizability of the findings.

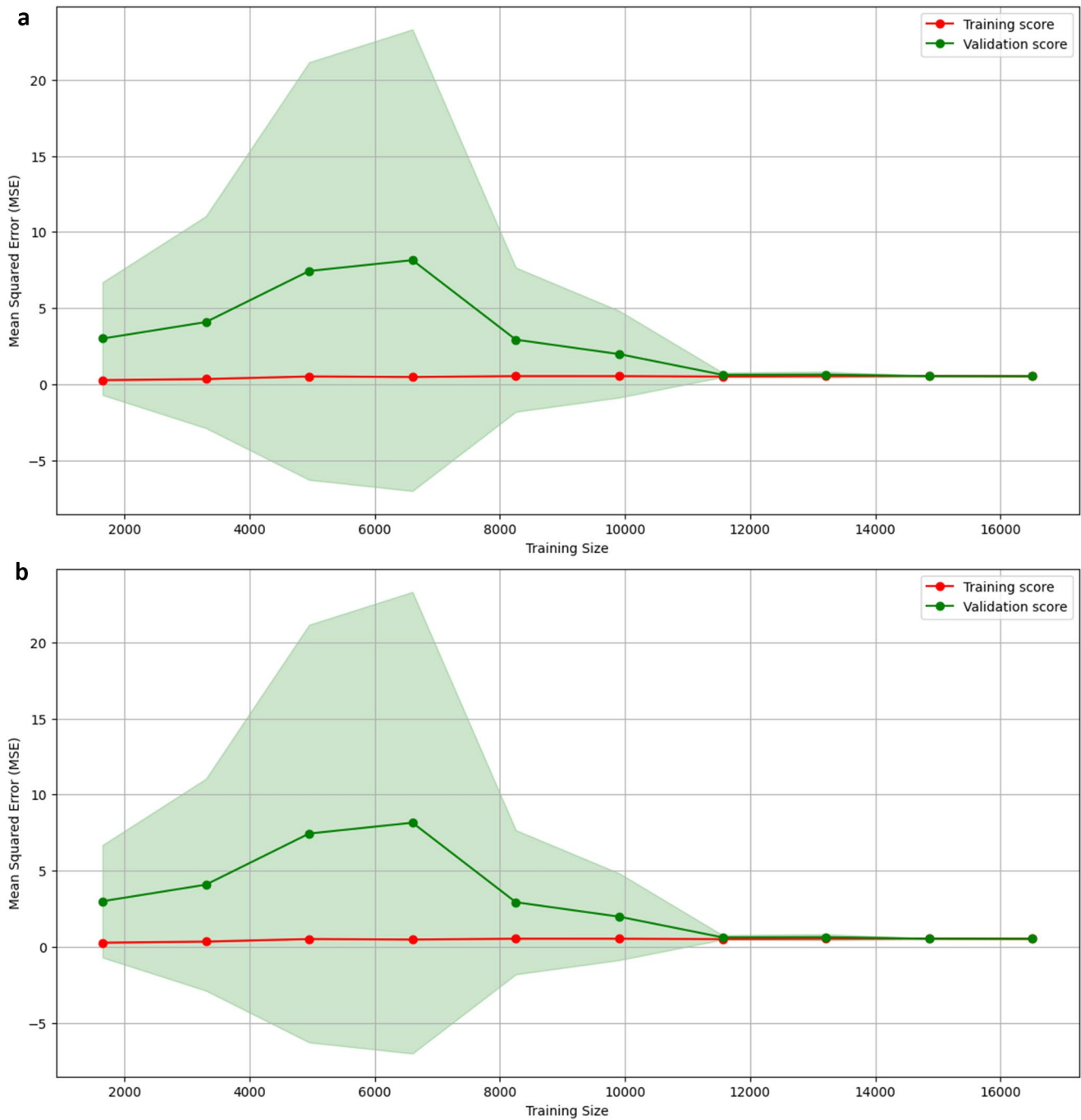


Fig. 11 (a) Learning curve- Linear regression. (b) Learning curve-Ridge regression

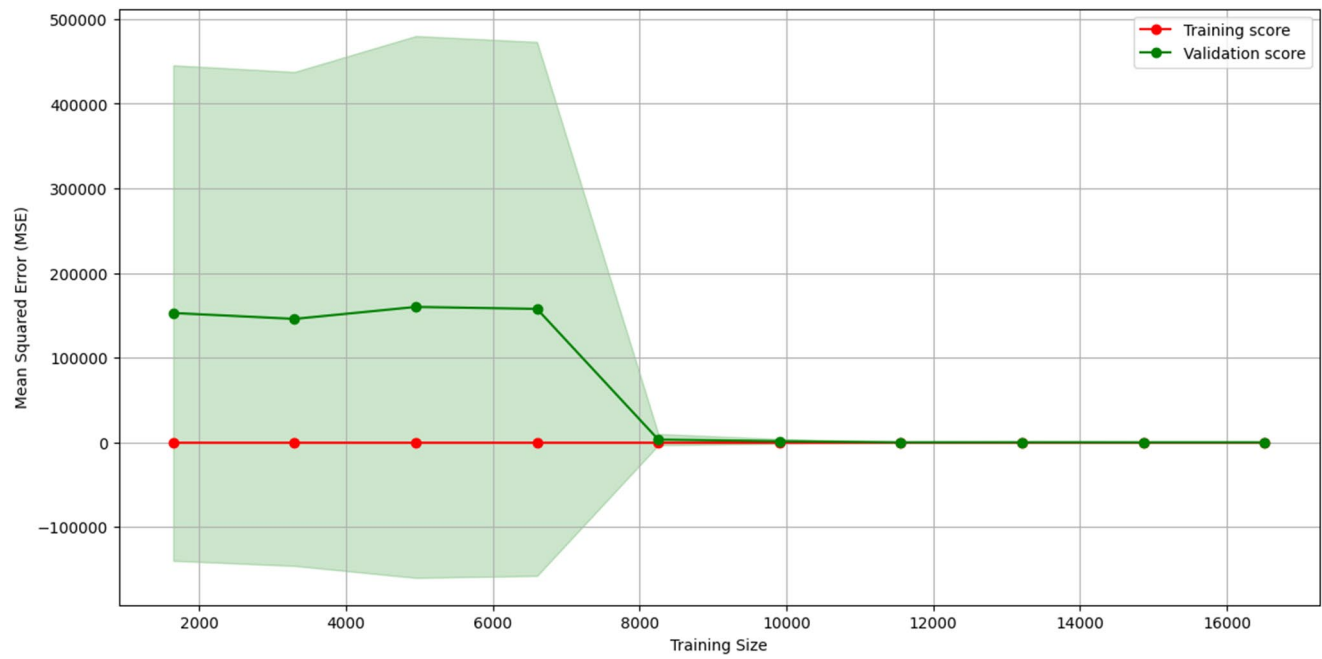


Fig. 12 Learning curve- polynomial Ridge regression

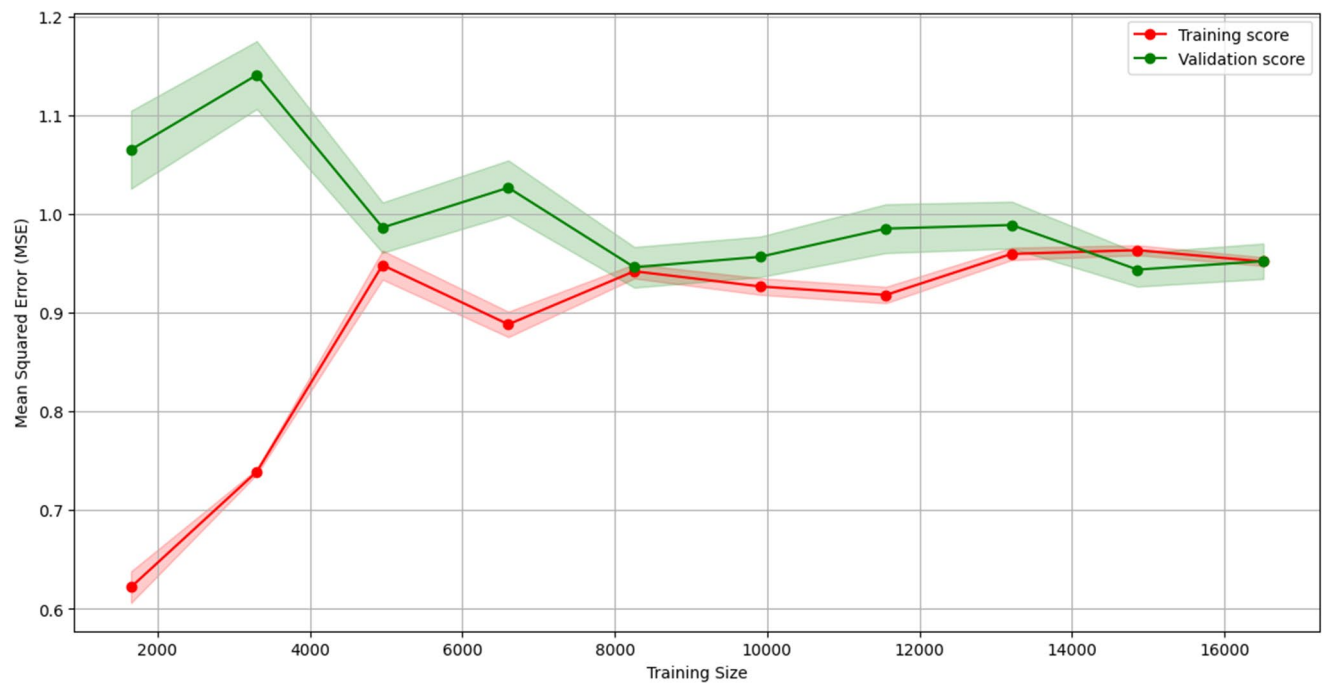


Fig. 13 Learning curve- Lasso regression

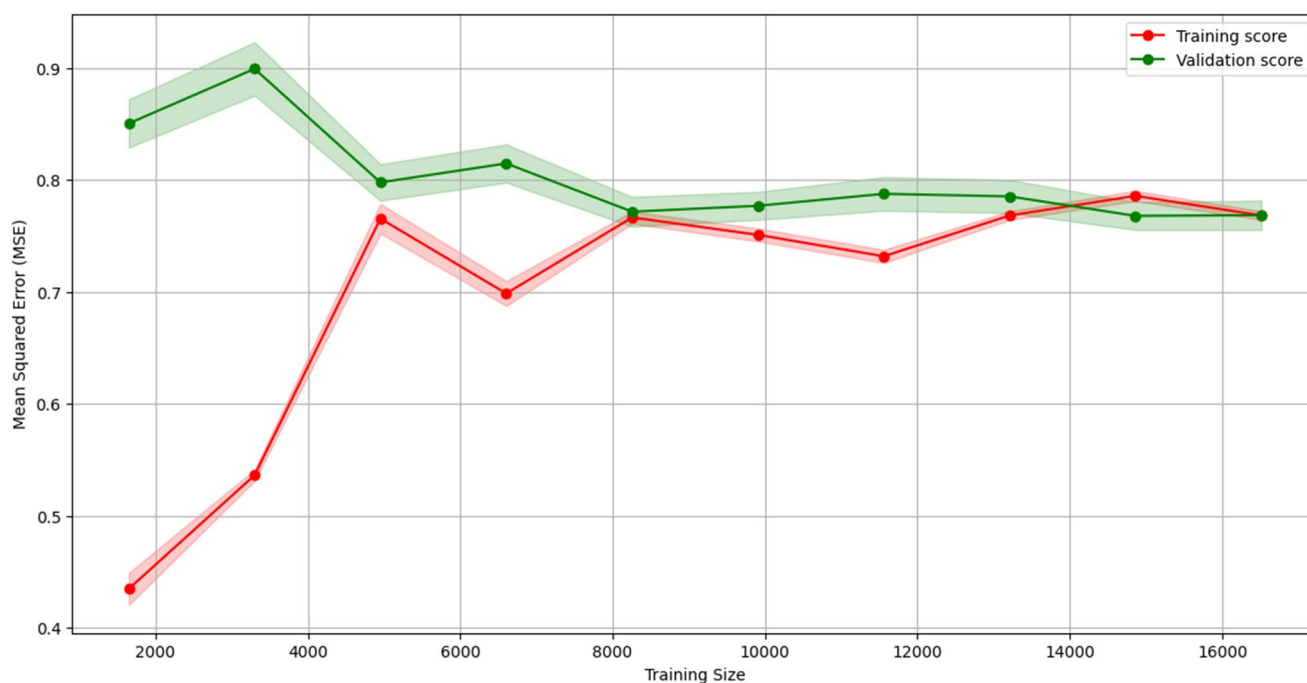


Fig. 14 Learning curve- elastic net regression

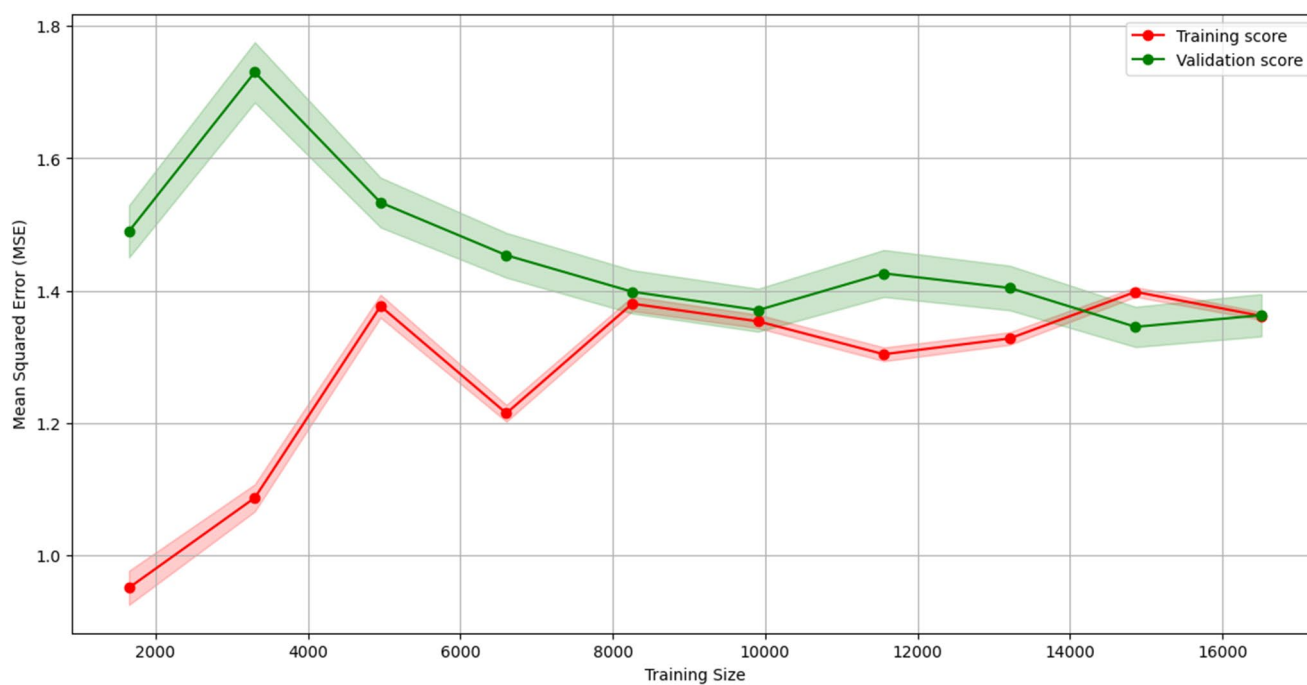


Fig. 15 Learning curve- SVR with RBF

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Data Availability The dataset produced and examined in this study can be obtained upon reasonable request from the corresponding author.

Author Contributions All authors participated equally in this research.

Declarations

Conflict of Interest There is no conflict of interest in this research work.

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